

Controlled Distillation of Speculative LLM Venture Concepts:

Evaluator-Assessed Evidence for Bounded MVP Hypotheses

Anonymous author(s) for review

Abstract

Large language models can generate unusual venture concepts, but speculative outputs often combine a potentially distinctive mechanism with unsupported claims, broad dependencies, and an MVP scope that is difficult to test. We introduce Controlled Hallucination Venture Discovery (CHVD), a controlled concept-distillation procedure that separates speculative venture-concept generation (M1) from a bounded transformation (M3). The objective is not to establish market success or real-world startup readiness. Instead, the study tests whether a transformation can retain a concept’s distinctive creative mechanism while improving evaluator-assessed prototype readiness at the level of written concept cards. The primary study uses 60 matched M1–M3 pairs from Gemini 3.1 Flash Lite P1 across six domains. Two independent LLM evaluators performed blinded individual-card scoring, and two role-aware directional audits assessed creative-core preservation and practical MVP improvement. In individual scoring, technical plausibility was higher for M3 than for M1 for both evaluators after Holm correction (ChatGPT: $\Delta = +0.150$, $p_{\text{Holm}} = .025$; Gemini: $\Delta = +0.433$, $p_{\text{Holm}} < .001$). Directional audits rated creative-core preservation near 4/5, practical MVP improvement above 4.27/5, and overclaim-removal quality above 4/5 for both evaluators. The results support a bounded claim: CHVD can function as an auditable procedure for converting selected speculative mechanisms into more testable MVP hypotheses. They do not demonstrate human preference, factual correctness, market viability, startup success, or universal superiority of transformed concepts.

Keywords: large language models; venture ideation; concept distillation; hallucination; prototype readiness; LLM-as-a-judge; minimum viable product

1 Introduction

Large language models (LLMs) can generate plausible and unusual venture concepts. The same capacity for non-obvious association can also introduce unsupported technical claims, unclear dependencies, and scopes too broad to test. In early ideation, the question is not whether unsupported content is intrinsically valuable. It is whether a controlled process can retain a distinctive creative mechanism while converting the mechanism into a constrained experimental hypothesis.

We introduce Controlled Hallucination Venture Discovery (CHVD), a two-stage within-model concept-distillation framework. M1 is a speculative venture concept that may contain a distinctive but unverified mechanism. M3 is a controlled, bounded reformulation intended to preserve that mechanism while replacing open-ended dependencies or unsupported assumptions with a testable MVP hypothesis.

The contribution is deliberately narrow. We report evaluator-assessed properties of written cards rather than market outcomes. We also separate individual-card scoring, role-aware directional traceability, and neutral paired comparison because those protocols answer different questions and cannot be treated as interchangeable evidence.

2 Study design and evidence policy

Table 1 distinguishes the primary Gemini 3.1 Flash Lite P1 study from earlier Gemini 2.5 Flash Lite P0/P1 developmental studies. The studies differ in corpus construction, domains, prompt regimes, and evaluation sessions; raw scores are not pooled and no generator-model ranking is claimed.

Table 1: Study inventory and evidential role.

Study	Pairs	Role	Individual/directional evidence	Neutral paired status
Gemini 2.5 FL P0	38	Supporting	Developmental only; not pooled	Not used for main inference
Gemini 2.5 FL P1	35	Supporting	Developmental only; not pooled	Non-confirmatory after Holm
Gemini 3.1 FL P1	60	Primary	Main individual and directional evidence	Quarantined; not used for efficacy/equivalence

The primary individual protocol compares matched M1–M3 cards for ten metrics using paired Wilcoxon signed-rank tests, with Holm correction within each evaluator. The cross-evaluator replication rule is strict: a result is called replicated only when the direction is compatible and the Holm-adjusted test is significant for both evaluators. Risk metrics are oriented so lower values favor M3.

The directional protocol is intentionally role-aware: evaluators see the reference and transformation labels because creative-core preservation and semantic drift cannot be assessed without that relationship. Directional prototype choice therefore measures traceability-oriented recommendation, not neutral A/B preference.

3 Results

3.1 All individual metrics in the primary study

Table 2 reports every primary-study individual metric, not only the significant findings. Technical plausibility is the only result satisfying the pre-specified cross-evaluator replication criterion. Gemini additionally reports M3-favorable changes in MVP clarity, unsupported-claim risk, ethical/privacy risk, overall MVP usefulness, and prototype willingness; these remain evaluator-specific secondary findings. The full statistics, bootstrap intervals, and tie counts are in Supplementary Table S1 and the machine-readable file `results/tables/S01_Gemini31_P1_Individual_All_Metrics.csv`.

Table 2: Gemini 3.1 Flash Lite P1: all individual-card metrics ($n = 60$ pairs per evaluator). $\Delta = \text{M3} - \text{M1}$. “Rep.” requires Holm significance for both evaluators in the same direction.

Metric	GPT Δ	GPT p_H	Gemini Δ	Gemini p_H	Rep.
Novelty	-0.067	1.000	-0.217	.010	No
Conceptual Coherence	+0.042	1.000	+0.067	.656	No
User Pain Plausibility	+0.000	—	+0.033	1.000	No
Technical Plausibility	+0.150	.025	+0.433	<.001	Yes
MVP Clarity	-0.025	1.000	+0.583	<.001	No
Business Model Plausibility	+0.133	.163	-0.033	1.000	No
Unsupported Claim Risk	-0.192	.110	-0.333	<.001	No
Ethical Privacy Risk	-0.033	1.000	-0.117	.062	No
Overall MVP Usefulness	+0.050	.089	+0.233	.014	No
Prototype Willingness	+0.055	.110	+0.367	<.001	No

3.2 Directional traceability evidence

Table 3 reports all five directional metrics. Both evaluators rated creative-core preservation near 4/5, practical MVP improvement above 4.27/5, and overclaim-removal quality above 4/5. Semantic-drift risk was below the midpoint for both evaluators. This is the main evidence about the intended process mechanism: preservation of a recognizable creative core alongside practical bounding.

Table 3: Gemini 3.1 Flash Lite P1: role-aware directional metrics ($n = 60$ per evaluator).

Metric	GPT mean	GPT 95% CI	Gemini mean	Gemini 95% CI
Creative Core Preservation	3.998	[3.893, 4.105]	4.030	[3.868, 4.182]
Novelty Retention	3.803	[3.692, 3.917]	3.408	[3.230, 3.582]
Practical MVP Improvement	4.370	[4.325, 4.415]	4.277	[4.182, 4.370]
Overclaim Removal Quality	4.100	[4.032, 4.168]	4.077	[3.962, 4.188]
Semantic Drift Risk	2.428	[2.273, 2.580]	2.013	[1.850, 2.187]

Table 4: Gemini 3.1 Flash Lite P1: role-aware directional prototype choice. This is not a neutral preference test.

Evaluator	Transformation	Reference	Tie	Transform. share excl. ties	Exact p
ChatGPT Plus 5.5 Web	48	12	0	.800	3.18×10^{-6}
Gemini 3.1 Pro Web	37	5	18	.881	4.43×10^{-7}

3.3 Neutral paired evidence is retained but not used for efficacy inference

The repository retains detailed neutral paired tables for transparency. Their evidential status is reported in Table 5. Gemini 3.1 P1 neutral outputs are quarantined from efficacy/equivalence inference because later diagnostics identified position-sensitive behavior; Gemini neutral v1/v2/v3 outputs are also archived as excluded preference-consistency artifacts. Gemini 2.5 P1 neutral results are reported in the supplement but are non-confirmatory after Holm correction. Consequently, no main paper claim relies on neutral paired results.

Table 5: Evidence disposition.

Evidence stream	Repository treatment	Manuscript use
G3.1 individual	Fully reported, strict schema audits pass	Primary individual inference
G3.1 directional	Fully reported, strict schema audits pass	Primary traceability inference
G3.1 neutral	Full archive retained; quarantined	No efficacy/equivalence claim
G2.5 P0/P1	Full detailed supplement retained	Supporting/developmental only
Human evaluation	Protocol prepared, no outcomes collected	No claim

4 Discussion

The evidence supports a limited interpretation of CHVD. A speculative concept can serve as a source of a distinctive creative mechanism. A controlled transformation can then express that mechanism through a smaller, more testable MVP hypothesis. The most reliable individual-card

effect in the primary corpus is higher evaluator-assessed technical plausibility. The strongest process evidence is directional: substantial creative-core preservation, high practical-MVP improvement, high overclaim-removal quality, and lower semantic-drift risk.

The results indicate a structured novelty–practicalization trade-off rather than an unconditional quality gain. The study does not establish factual truth, implementation success, user adoption, profitability, or market viability.

5 Limitations and reproducibility

All reported outcomes are LLM-evaluator assessments of written concept cards. They are not human or expert judgments and are not implementation outcomes. A blinded human evaluation is planned as external validation. The repository includes this manuscript, its compiled PDF, all aggregate result tables, source outputs, validators, hashes, and explicit exclusions. It does not release private A/B keys, candidate-level unblinding maps, credentials, account/session metadata, or human-participant data.

6 Conclusion

Within the evaluated Gemini 3.1 Flash Lite P1 matched corpus, CHVD provides auditable evidence for a constrained transformation from selected speculative mechanisms to evaluator-assessed testable MVP hypotheses. The findings do not claim startup readiness. They support a narrower claim of controlled LLM concept distillation, with human evaluation as the next external validation layer.

References

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